```
# ============================================================
# IMPORT LIBRARIES
# ============================================================

import warnings
warnings.filterwarnings("ignore")

import numpy as np
import pandas as pd

import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split

from sklearn.preprocessing import (
    LabelEncoder,
    OneHotEncoder,
    StandardScaler
)
```

```
# ============================================================
# LOAD DATASET
# ============================================================

df = pd.read_csv("/kaggle/input/datasets/blastchar/telco-customer-churn/WA_Fn-UseC_-Telco-Customer-Churn.csv")

print(df.shape)

df.head()
```

```
(7043, 21)
```

	customerID	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	MultipleLines	InternetService	OnlineSecurity	...	DeviceProtectio
0	7590-VHVEG	Female	0	Yes	No	1	No	No phone service	DSL	No	...	N
1	5575-GNVDE	Male	0	No	No	34	Yes	No	DSL	Yes	...	Ye
2	3668-QPYBK	Male	0	No	No	2	Yes	No	DSL	Yes	...	N
3	7795-CFOCW	Male	0	No	No	45	No	No phone service	DSL	Yes	...	Ye
4	9237-HQITU	Female	0	No	No	2	Yes	No	Fiber optic	No	...	N

5 rows × 21 columns

```
import os

for dirname, _, filenames in os.walk('/kaggle/input'):
    for filename in filenames:
        print(os.path.join(dirname, filename))
```

```
/kaggle/input/datasets/blastchar/telco-customer-churn/WA_Fn-UseC_-Telco-Customer-Churn.csv
```

```
# ============================================================
# BASIC INFORMATION
# ============================================================

print(df.info())

print("\n")

print(df.describe(include="all"))

print("\n")

print(df.dtypes)
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7043 entries, 0 to 7042
Data columns (total 21 columns):
 #   Column            Non-Null Count  Dtype
---  ------            --------------  -----
 0   customerID        7043 non-null   object
 1   gender            7043 non-null   object
 2   SeniorCitizen     7043 non-null   int64
 3   Partner           7043 non-null   object
 4   Dependents        7043 non-null   object
 5   tenure            7043 non-null   int64
 6   PhoneService      7043 non-null   object
 7   MultipleLines     7043 non-null   object
 8   InternetService   7043 non-null   object
 9   OnlineSecurity    7043 non-null   object
 10  OnlineBackup      7043 non-null   object
 11  DeviceProtection  7043 non-null   object
 12  TechSupport       7043 non-null   object
 13  StreamingTV       7043 non-null   object
 14  StreamingMovies   7043 non-null   object
 15  Contract          7043 non-null   object
 16  PaperlessBilling  7043 non-null   object
 17  PaymentMethod     7043 non-null   object
```

```
 18  MonthlyCharges    7043 non-null   float64
 19  TotalCharges      7043 non-null   object
 20  Churn             7043 non-null   object
dtypes: float64(1), int64(2), object(18)
memory usage: 1.1+ MB
None


        customerID gender  SeniorCitizen Partner Dependents       tenure  \
count         7043   7043    7043.000000    7043       7043  7043.000000
unique        7043      2            NaN       2          2          NaN
top     7590-VHVEG   Male            NaN      No         No          NaN
freq             1   3555            NaN    3641       4933          NaN
mean           NaN    NaN       0.162147     NaN        NaN    32.371149
std            NaN    NaN       0.368612     NaN        NaN    24.559481
min            NaN    NaN       0.000000     NaN        NaN     0.000000
25%            NaN    NaN       0.000000     NaN        NaN     9.000000
50%            NaN    NaN       0.000000     NaN        NaN    29.000000
75%            NaN    NaN       0.000000     NaN        NaN    55.000000
max            NaN    NaN       1.000000     NaN        NaN    72.000000

       PhoneService MultipleLines InternetService OnlineSecurity  ...  \
count          7043          7043            7043           7043  ...
unique            2             3               3              3  ...
top             Yes            No     Fiber optic             No  ...
freq           6361          3390            3096           3498  ...
mean            NaN           NaN             NaN            NaN  ...
std             NaN           NaN             NaN            NaN  ...
min             NaN           NaN             NaN            NaN  ...
25%             NaN           NaN             NaN            NaN  ...
50%             NaN           NaN             NaN            NaN  ...
75%             NaN           NaN             NaN            NaN  ...
max             NaN           NaN             NaN            NaN  ...

       DeviceProtection TechSupport StreamingTV StreamingMovies  \
```

```
# ============================================================
# DATASET OVERVIEW
# ============================================================

print("Rows :", df.shape[0])

print("Columns :", df.shape[1])

print("Duplicate Rows :", df.duplicated().sum())

print("Missing Values :")

display(df.isnull().sum())
```

```
Rows : 7043
Columns : 21
Duplicate Rows : 0
Missing Values :
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

```
# ============================================================
# CHECK MISSING VALUES
# ============================================================

missing = (
    df.isnull()
      .sum()
      .sort_values(ascending=False)
)

display(missing)
```

```
customerID          0
gender              0
SeniorCitizen       0
Partner             0
Dependents          0
tenure              0
PhoneService        0
MultipleLines       0
InternetService     0
OnlineSecurity      0
OnlineBackup        0
DeviceProtection    0
TechSupport         0
StreamingTV         0
StreamingMovies     0
Contract            0
PaperlessBilling    0
PaymentMethod       0
MonthlyCharges      0
TotalCharges        0
Churn               0
dtype: int64
```

```
# ============================================================
# UNIQUE VALUES
# ============================================================

for col in df.columns:

    print("="*50)

    print(col)

    print(df[col].nunique())

    print(df[col].unique()[:10])
```

```
==================================================
customerID
7043
['7590-VHVEG' '5575-GNVDE' '3668-QPYBK' '7795-CFOCW' '9237-HQITU'
 '9305-CDSKC' '1452-KIOVK' '6713-OKOMC' '7892-POOKP' '6388-TABGU']
==================================================
gender
2
['Female' 'Male']
==================================================
SeniorCitizen
2
[0 1]
==================================================
Partner
2
['Yes' 'No']
==================================================
Dependents
2
['No' 'Yes']
==================================================
tenure
73
[ 1 34  2 45  8 22 10 28 62 13]
==================================================
PhoneService
2
['No' 'Yes']
==================================================
MultipleLines
3
['No phone service' 'No' 'Yes']
==================================================
InternetService
3
['DSL' 'Fiber optic' 'No']
==================================================
OnlineSecurity
3
['No' 'Yes' 'No internet service']
==================================================
OnlineBackup
3
['Yes' 'No' 'No internet service']
==================================================
DeviceProtection
3
['No' 'Yes' 'No internet service']
==================================================
TechSupport
3
['No' 'Yes' 'No internet service']
==================================================
StreamingTV
3
['No' 'Yes' 'No internet service']
==================================================
```

```
# ============================================================
# COLUMN TYPES
# ============================================================

numeric_cols = df.select_dtypes(include=np.number).columns.tolist()
```

```
categorical_cols = df.select_dtypes(include="object").columns.tolist()

print("Numerical Features")

print(numeric_cols)

print()

print("Categorical Features")

print(categorical_cols)
```

```
Numerical Features
['SeniorCitizen', 'tenure', 'MonthlyCharges']

Categorical Features
['customerID', 'gender', 'Partner', 'Dependents', 'PhoneService', 'MultipleLines', 'InternetService', 'OnlineSecurity', 'OnlineBackup', 'DeviceProte
```

```
# Check data type
print(df["TotalCharges"].dtype)

# Count blank strings
print((df["TotalCharges"] == " ").sum())
```

```
object
11
```

```
df.drop(columns=["customerID"], inplace=True)
```

```
df["TotalCharges"] = df["TotalCharges"].replace(" ", np.nan)
```

```
df["TotalCharges"] = pd.to_numeric(df["TotalCharges"])
```

```
df["TotalCharges"].isnull().sum()
```

```
np.int64(11)
```

```
df.dropna(inplace=True)
```

```
df.reset_index(drop=True, inplace=True)
```

```
print(df.shape)
print(df.isnull().sum().sum())
```

```
(7032, 20)
0
```

```
for col in df.select_dtypes(include="object").columns:
    print(f"\n{col}")
    print(df[col].value_counts())
```

```

gender
gender
Male      3549
Female    3483
Name: count, dtype: int64

Partner
Partner
No     3639
Yes    3393
Name: count, dtype: int64

Dependents
Dependents
No     4933
Yes    2099
Name: count, dtype: int64

PhoneService
PhoneService
Yes    6352
No      680
Name: count, dtype: int64

MultipleLines
MultipleLines
No                  3385
Yes                 2967
No phone service     680
Name: count, dtype: int64

InternetService
InternetService
Fiber optic    3096
DSL            2416
No             1520
Name: count, dtype: int64

OnlineSecurity
OnlineSecurity
No                     3497
Yes                    2015
No internet service    1520
Name: count, dtype: int64
```

```
OnlineBackup
OnlineBackup
No                     3087
Yes                    2425
No internet service    1520
Name: count, dtype: int64

DeviceProtection
DeviceProtection
No                     3094
Yes                    2418
No internet service    1520
```

```
# ============================================================
# FEATURE ENGINEERING
# ============================================================

# ------------------------------------------------------------
# 1. Total Services
# Count the number of subscribed services
# ------------------------------------------------------------

service_columns = [
    "PhoneService",
    "MultipleLines",
    "OnlineSecurity",
    "OnlineBackup",
    "DeviceProtection",
    "TechSupport",
    "StreamingTV",
    "StreamingMovies"
]

df["TotalServices"] = 0

for col in service_columns:
    df["TotalServices"] += df[col].isin(["Yes"]).astype(int)

print("✓ TotalServices created")
```

```
✓ TotalServices created
```

```
# ------------------------------------------------------------
# 2. Customer Lifetime Value (Proxy)
# ------------------------------------------------------------

df["LifetimeValue"] = (
    df["MonthlyCharges"] *
    df["tenure"]
)

print("✓ LifetimeValue created")
```

```
✓ LifetimeValue created
```

```
# ------------------------------------------------------------
# 3. Average Monthly Spend
# ------------------------------------------------------------

df["AvgMonthlySpend"] = (
    df["TotalCharges"] /
    df["tenure"]
)

print("✓ AvgMonthlySpend created")
```

```
✓ AvgMonthlySpend created
```

```
# ------------------------------------------------------------
# 4. Internet User
# ------------------------------------------------------------

df["HasInternet"] = (
    df["InternetService"] != "No"
).astype(int)

print("✓ HasInternet created")
```

```
✓ HasInternet created
```

```
# ------------------------------------------------------------
# 5. Protection Score
# ------------------------------------------------------------

protection_columns = [
    "OnlineSecurity",
    "DeviceProtection",
    "TechSupport"
]

df["ProtectionScore"] = 0

for col in protection_columns:
    df["ProtectionScore"] += (
        df[col] == "Yes"
```

```
    ).astype(int)

print("✓ ProtectionScore created")
```

```
✓ ProtectionScore created
```

```
# Base services
df["TotalServices"] = (
    (df["PhoneService"] == "Yes").astype(int)
    + (df["MultipleLines"] == "Yes").astype(int)
    + (df["InternetService"] != "No").astype(int)
    + (df["OnlineSecurity"] == "Yes").astype(int)
    + (df["OnlineBackup"] == "Yes").astype(int)
    + (df["DeviceProtection"] == "Yes").astype(int)
    + (df["TechSupport"] == "Yes").astype(int)
    + (df["StreamingTV"] == "Yes").astype(int)
    + (df["StreamingMovies"] == "Yes").astype(int)
)
```

```
# ============================================================
# LABEL ENCODING
# ============================================================

binary_mapping = {
    "Yes": 1,
    "No": 0,
    "Male": 1,
    "Female": 0
}

binary_columns = [
    "gender",
    "Partner",
    "Dependents",
    "PhoneService",
    "PaperlessBilling",
    "Churn"
]

for col in binary_columns:
    df[col] = df[col].map(binary_mapping)

print("✓ Binary columns encoded")
```

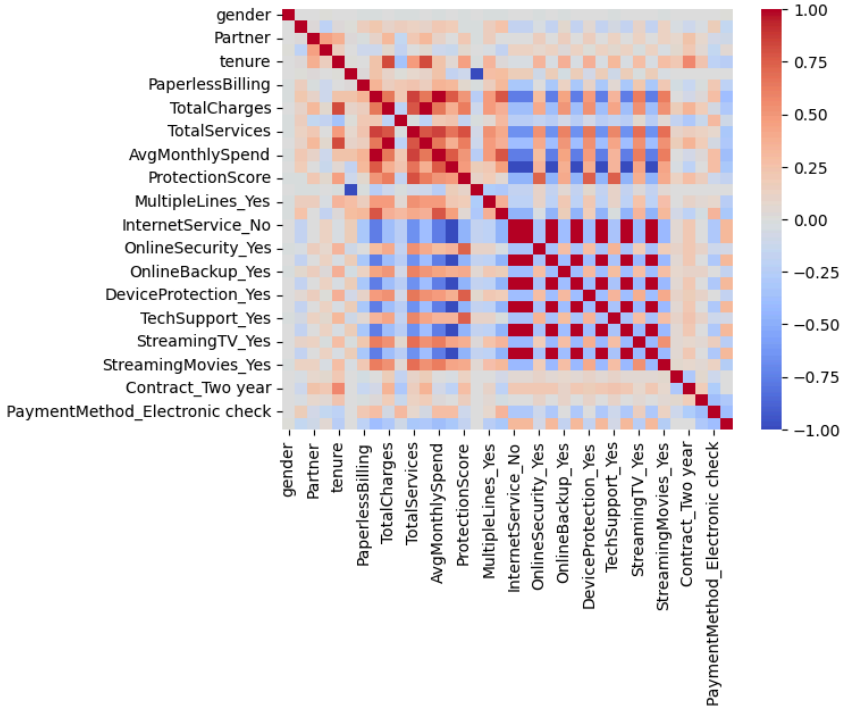
```
✓ Binary columns encoded
```

```
import matplotlib.pyplot as plt
import seaborn as sns


corr = df.corr(numeric_only=True)

sns.heatmap(corr, cmap="coolwarm")

# Show the plot
plt.show()
```




```
# ============================================================
# ONE-HOT ENCODING
# ============================================================
```

```
multi_columns = [
    "MultipleLines",
    "InternetService",
    "OnlineSecurity",
    "OnlineBackup",
    "DeviceProtection",
    "TechSupport",
    "StreamingTV",
    "StreamingMovies",
    "Contract",
    "PaymentMethod"
]

df = pd.get_dummies(
    df,
    columns=multi_columns,
    drop_first=True,
    dtype=int
)

print("✓ One-Hot Encoding completed")
```

```
✓ One-Hot Encoding completed
```

```
print(df.shape)

df.head()
```

```
(7032, 36)
```

	gender	SeniorCitizen	Partner	Dependents	tenure	PhoneService	PaperlessBilling	MonthlyCharges	TotalCharges	Churn	...	TechSupport_Yes	Stre
0	0	0	1	0	1	0	1	29.85	29.85	0	...	0	
1	1	0	0	0	34	1	0	56.95	1889.50	0	...	0	
2	1	0	0	0	2	1	1	53.85	108.15	1	...	0	
3	1	0	0	0	45	0	0	42.30	1840.75	0	...	1	
4	0	0	0	0	2	1	1	70.70	151.65	1	...	0	

5 rows × 36 columns

```
df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 7032 entries, 0 to 7031
Data columns (total 36 columns):
 #   Column                                 Non-Null Count  Dtype
---  ------                                 --------------  -----
 0   gender                                 7032 non-null   int64
 1   SeniorCitizen                          7032 non-null   int64
 2   Partner                                7032 non-null   int64
 3   Dependents                             7032 non-null   int64
 4   tenure                                 7032 non-null   int64
 5   PhoneService                           7032 non-null   int64
 6   PaperlessBilling                       7032 non-null   int64
 7   MonthlyCharges                         7032 non-null   float64
 8   TotalCharges                           7032 non-null   float64
 9   Churn                                  7032 non-null   int64
 10  TotalServices                          7032 non-null   int64
 11  LifetimeValue                          7032 non-null   float64
 12  AvgMonthlySpend                        7032 non-null   float64
 13  HasInternet                            7032 non-null   int64
 14  ProtectionScore                        7032 non-null   int64
 15  MultipleLines_No phone service         7032 non-null   int64
 16  MultipleLines_Yes                      7032 non-null   int64
 17  InternetService_Fiber optic            7032 non-null   int64
 18  InternetService_No                     7032 non-null   int64
 19  OnlineSecurity_No internet service     7032 non-null   int64
 20  OnlineSecurity_Yes                     7032 non-null   int64
 21  OnlineBackup_No internet service       7032 non-null   int64
 22  OnlineBackup_Yes                       7032 non-null   int64
 23  DeviceProtection_No internet service   7032 non-null   int64
 24  DeviceProtection_Yes                   7032 non-null   int64
 25  TechSupport_No internet service        7032 non-null   int64
 26  TechSupport_Yes                        7032 non-null   int64
 27  StreamingTV_No internet service        7032 non-null   int64
 28  StreamingTV_Yes                        7032 non-null   int64
 29  StreamingMovies_No internet service    7032 non-null   int64
 30  StreamingMovies_Yes                    7032 non-null   int64
 31  Contract_One year                      7032 non-null   int64
 32  Contract_Two year                      7032 non-null   int64
 33  PaymentMethod_Credit card (automatic)  7032 non-null   int64
 34  PaymentMethod_Electronic check         7032 non-null   int64
 35  PaymentMethod_Mailed check             7032 non-null   int64
dtypes: float64(4), int64(32)
memory usage: 1.9 MB
```

```
print(df["Churn"].value_counts())

print(df["Churn"].value_counts(normalize=True) * 100)
```

```
Churn
0    5163
1    1869
```

```
Name: count, dtype: int64
Churn
0    73.421502
1    26.578498
Name: proportion, dtype: float64
```

```
# ============================================================
# SAVE PREPROCESSED DATASET
# ============================================================

print(df.head(10))
df.to_csv("telco_preprocessed.csv", index=False)

print("✓ Preprocessed dataset saved successfully!")
```

```
   gender  SeniorCitizen  Partner  Dependents  tenure  PhoneService  \
0       0              0        1           0       1             0
1       1              0        0           0      34             1
2       1              0        0           0       2             1
3       1              0        0           0      45             0
4       0              0        0           0       2             1
5       0              0        0           0       8             1
6       1              0        0           1      22             1
7       0              0        0           0      10             0
8       0              0        1           0      28             1
9       1              0        0           1      62             1

   PaperlessBilling  MonthlyCharges  TotalCharges  Churn  ...  \
0                 1           29.85         29.85      0  ...
1                 0           56.95       1889.50      0  ...
2                 1           53.85        108.15      1  ...
3                 0           42.30       1840.75      0  ...
4                 1           70.70        151.65      1  ...
5                 1           99.65        820.50      1  ...
6                 1           89.10       1949.40      0  ...
7                 0           29.75        301.90      0  ...
8                 1          104.80       3046.05      1  ...
9                 0           56.15       3487.95      0  ...

   TechSupport_Yes  StreamingTV_No internet service  StreamingTV_Yes  \
0                0                                0                0
1                0                                0                0
2                0                                0                0
3                1                                0                0
4                0                                0                0
5                0                                0                1
6                0                                0                1
7                0                                0                0
8                1                                0                1
9                0                                0                0

   StreamingMovies_No internet service  StreamingMovies_Yes  \
0                                    0                    0
1                                    0                    0
2                                    0                    0
3                                    0                    0
4                                    0                    0
5                                    0                    1
6                                    0                    0
7                                    0                    0
8                                    0                    1
9                                    0                    0

   Contract_One year  Contract_Two year  \
0                  0                  0
1                  1                  0
2                  0                  0
3                  1                  0
4                  0                  0
5                  0                  0
6                  0                  0
7                  0                  0
8                  0                  0
```

```
Start coding or generate with AI.
```